
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-
2 racy. Determining how much normally requires re-running a full benchmark suite,
3 which is the expense quantization was meant to avoid. We ask whether the accu-
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38
5 base checkpoints, 8 model families and 6 quantization schemes, we report three
6 results. First, a *negative* one: per-model prediction carries almost no signal. Un-
7 der leave-one-family-out validation the best predictor, a per-scheme mean, beats
8 a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]),
9 and both ridge regression and gradient boosting are *worse* than guessing the av-
10 erage, because evaluation noise accounts for at least a third of the variance on 8
11 of 11 benchmarks. Second, a positive one: conditioning only on the quantization
12 scheme yields split-conformal intervals with 90.1% empirical coverage (118/131,
13 95% CI [83.6, 94.6]) on checkpoints never used in construction. Those check-
14 points cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131 rows. Each
15 (scheme, size-band) cell returns one of three verdicts: trusted; insufficient evi-
16 dence, where the interval is still shown but the scheme is demoted and flagged;
17 or refused, where the interval is withheld. Currently 8 of 17 cells are insufficient
18 evidence and 0 are refused. Third, and most consequential for practice: the pub-
19 lished corpus is outcome-truncated. Every configuration in it is one that someone
20 chose to release. We rented an A100 and measured deliberately-faulted configu-
21 rations, observing losses to -39.5pp against a published worst case of -8.86pp —
22 measured, we stress, only on models of 1.5B and below, and on configurations
23 engineered to fail rather than typical ones. This is an existence proof that far
24 worse outcomes exist than the corpus contains; it does not estimate how common
25 they are. We argue the resulting artifact should be described as a calibrated his-
26 torical baseline rather than a predictor, and we state precisely why its coverage
27 guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and
28 the provenance of every reported figure are provided as supplementary material.

29 1 Introduction

30 Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8
31 wants one number: how much accuracy will this cost? In practice the only way to know is to run the
32 evaluation.

33 This paper asks whether that number can be forecast from the public record. Model publishers
34 release quantized checkpoints with benchmark tables comparing the compressed model against its
35 dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of
36 interest (850 extracted in our harvest).

37 Our contributions:

- 38 1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and
39 we show why: the target is dominated by evaluation noise. We report this as the primary
40 finding rather than burying it.
- 41 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured
42 coverage on held-out families and a three-state verdict for each (scheme, size-band) cell:
43 *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted
44 and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld.
45 Currently 8 of 17 cells are insufficient evidence and 0 are refused.
- 46 3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on
47 deliberately-faulted configurations—data the published record structurally cannot contain.
48 It shows that losses far beyond anything published are reachable on small models. It does
49 not estimate how large the bias is: our attempt at a correction (in the supplement) did not
50 work well enough to ship.
- 51 4. **An audit** (§9) recording errors we found in our own work, including ones that changed
52 published numbers and claims.

53 2 Related Work

54 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism
55 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and
56 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks
57 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is
58 a model’s absolute score; ours is the paired delta between a model and its compressed self, and
59 quantized checkpoints do not appear in their paper.

60 **Conformal prediction on compressed models.** Tong et al. [2026] apply conformal prediction
61 directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand
62 differs: their conformal sets are over label space, constructed from a compressed model’s own output
63 probabilities, so the method measures uncertainty *after* running the model. Ours is constructed over
64 historical accuracy deltas to give an interval *before* running anything. Their Figure 2 plots the
65 same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it. Their finding that compression
66 decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can claim
67 (§8).

68 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]
69 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024]. Kumar et al. [2024]
70 fit a functional form that predicts degradation from training and inference in varied precisions, using
71 pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per
72 scheme, over published post-training-quantized checkpoints.

73 **The source of our data.** Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-
74 3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested.
75 As far as we can tell, that work does not address whether the published set of quantized models is a
76 biased sample. Characterising that gap is the contribution of §7.

77 **Selection bias.** Survivorship bias has been identified and measured in a widely used retrieval
78 benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail un-
79 der selection: conditional on a selection event, calibration data are no longer exchangeable with a
80 test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

81 3 Data

82 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
83 ing 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near
84 chance, where a delta is uninformative), across 38 base checkpoints, 8 families and 6 schemes.

85 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
 86 This is redundant information, and we exploit it: for every three-column row we check recovery $\approx$
 87 $100 \times \text{after/before}$ to a tolerance derived from the card’s own printed precision, and reject rows that
 88 disagree. This caught spurious values that our parser produced under column-order variation be-
 89 tween card templates, and internally inconsistent cards. Rejections are recorded rather than silently
 90 dropped.

91 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
 92 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
 93 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

94 4 Method

95 **Split conformal.** Following split conformal regression Lei et al. [2018], given calibration residuals
 96 $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which gives
 97 exact finite-sample coverage under exchangeability. Where n is too small for that index to exist, we
 98 return ∞ rather than a misleadingly finite band.

99 **Mondrian (group-conditional) calibration.** Calibrating separately within categories is the Mon-
 100 drian construction of Vovk et al. [2005], applied to regression by Boström and Johansson [2020]. A
 101 single marginal interval advertised at 90% was in fact a 68% interval for NVFP4 users—the users
 102 who most need it. We therefore calibrate per scheme.

103 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 104 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 105 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 106 measurably poor on a sufficient number of scored rows. It is *insufficient evidence* when the cell rests
 107 on fewer than 3 distinct checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on
 108 its coverage straddles the 85% line. In that case the evidence cannot support either of the other two
 109 verdicts. The tool still prints the all-sizes interval for such a cell, but demotes the scheme to the
 110 lowest tier and states that the cell cannot be judged; only a refused cell has its interval withheld.

111 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 112 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one
 113 to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing
 114 a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the
 115 interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage
 116 across all scored rows is 95.1%.

117 Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and nei-
 118 ther is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 70.1%
 119 over 77 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. `w8a16|>10B` had
 120 81.1% over 217 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval, [47, 100]%,
 121 straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those two
 122 cells; applying the rule to every cell moved 8 of 17 into that state, including several whose point-
 123 estimate coverage looked comfortable. The one-sided scoring and the three-state rule were proposed
 124 by an external commenter who identifies as an AI account collaborating with the OpenLanguage-
 125 Model maintainers; we recomputed each of their figures from the raw rows before adopting either
 126 change, and one of their descriptions did not survive that check.

127 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not indepen-
 128 dent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct
 129 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

130 5 The negative result

131 Under leave-one-family-out validation, row-weighted MAE:

predictor	MAE (pp)	beats global mean?
per-scheme mean (shipped)	0.7227	yes
global mean (baseline)	0.7545	—
per-(scheme $\times$ benchmark)	0.7607	no
ridge regression	0.7639	no
per-benchmark mean	0.7647	no
gradient boosting	0.7878	no

The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of [0.01, 0.05]pp, and it is positive in 7 of the 8 held-out families. It is a distinguishable but modest gain. Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom each degraded out-of-family performance.

Why. Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39 pp. The target is a difference of two noisy quantities that largely cancel: the numerically smallest and hardest-to-predict summary of quantization damage available. Two cautions on the noise figures. They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise. And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which we attribute to sampling error rather than treat as a finding.

Implication. The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band, with no model and no conformal machinery, is a serious competitor: it looked better covered in-sample, though under the strict protocol that advantage disappears (§9). What conformal contributes is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

6 Results

scheme	90% interval (pp)	worst observed	% losing >3 pp	evals
fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

Prospective validation. After freezing the method we harvested checkpoints not used during development. Under a strict protocol that excludes any family present in training and any card whose format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI [83.6, 94.6], against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code with the pipeline reproduces this figure with zero disagreements. That checks the computation, not the validity of the design.

What the prospective set does and does not cover. The 131 strict rows come from 19 quantized checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95% interval of [83.2, 94.7]%, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8 and NVFP4 have no prospective rows, so the figure says nothing about them.

scheme	strict rows	checkpoints	covered (%)
w4a16	80	11	87.5
w8a8_int	36	5	91.7
fp8_dynamic	11	2	100.0
w8a16	4	1	100.0

164 W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%;
165 the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict
166 filter, coverage is 90.3%, so the filter did not manufacture the headline.

167 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme
168 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when
169 real damage occurs, the interval usually misses it. This is the single most important caveat in the
170 paper.

171 7 The measured left tail

172 The corpus contains only configurations someone chose to publish. To measure what it structurally
173 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

174 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
175 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
176 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
177 instead of per-group.

178 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
179 that catastrophic damage is reachable; it is not a measured bound for larger models.

180 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
181 reproduces published base accuracy (47.20 against 47.42; 59.57 against 60.98), closing the harness
182 question. It exposed a larger one: our own *correctly-configured* control is $2.6\times$ more damaging
183 than published W4A16 on closely comparable checkpoints (mean gap -3.78pp , $z = 2.9$ and 5.3).
184 We attribute this to approximately $171\times$ less calibration data than production defaults; we found
185 no bug. Consequently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus
186 production. It is a conservative anchor, not a bound, and we decline to publish a corrected point
187 estimate: the correction moves by 5.70pp depending on the anchor set, and we have only 6 rows
188 from two models.

189 8 Limitations

190 **The guarantee is conditional, and absent where it matters most.** Split conformal gives exact
191 coverage under exchangeability between calibration and test. Our calibration set is the set of config-
192 urations a publisher chose to release—a selection event. Conditional on selection, calibration points
193 are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al.
194 [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process.
195 It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have
196 released—which is the case a practitioner most wants covered.

197 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression frequently
198 decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A
199 scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none
200 of that.

201 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence
202 that practitioner and calibration volume can dominate scheme choice. We measured how confining
203 this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9
204 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What
205 the other publishers almost never provide is a *Recovery* column, which is what lets each row be
206 checked against $100\times$ after/before and rejected on disagreement: of their rows, 6 pass that gate,
207 all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because
208 others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones
209 published in a self-verifying form.

210 **What can and cannot be re-run.** The supplement recomputes every tagged figure from the data
 211 files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded
 212 results are included), the harvest from raw model cards (the cards are not redistributed), or the
 213 prospective-card checks (those files are not included), and re-running the low-rank comparison men-
 214 tioned in the supplement needs a checkout of BenchPress’s released code.

215 **Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run.
 216 We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over
 217 the 19 quantized checkpoints gives $[83.2, 94.7]\%$, close to the row-level interval, but the unit of
 218 analysis remains a row.

219 9 Audit

220 We record errors found in our own work, including those that changed published claims. This
 221 section exists because the alternative—reporting only what survived—would misrepresent how the
 222 result was obtained.

223 Errors that changed a published number.

- 224 • *A retracted headline.* An early claim that held-out models came from families never used
 225 in construction was wrong; they were new checkpoints of families already present. The
 226 strict figure is 118/131.
- 227 • *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was
 228 matched as Math-Lvl-5, contaminating 7 test rows with scores from a different bench-
 229 mark.
- 230 • *A centring mismatch.* Calibration measured residuals about the stratum mean while in-
 231 tervals centred on the scheme mean, mis-sizing every calibrated width. Found only by
 232 independent reimplementations.
- 233 • *A column-order fault.* One card template orders columns [recovery, base, quant], which our
 234 parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that
 235 card, passing only the one where base = quant made the check degenerate.
- 236 • *A retracted single-publisher claim.* We had asserted that one publisher was close to the
 237 only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation
 238 cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be
 239 verified (§8).
- 240 • *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-
 241 sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An
 242 external commenter asked for it to wrap the one-sided figure, which exposed the mismatch
 243 (§4).

244 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-
 245 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization
 246 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.
 247 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-
 248 sided empirical index requires more calibration points than are often available. An infinite interval
 249 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The
 250 change was reversed.

251 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without
 252 qualification. Checking the method against the conformal literature rather than against our own
 253 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected
 254 in §8.

255 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in
 256 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude
 257 the affected family from the strict headline.

Mechanical verification. The figures this paper reports as results are macros generated from a claims registry that recomputes each value from source artifacts, and the accompanying code fails a test if an outward-facing document contains an untraced numeric literal. The exceptions are historical figures in the error list above (row counts and the -68pp delta) and the 90% marginal-interval figure in §4, which are quoted from the provenance record rather than recomputed.

10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is largely measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints for 4 of the 6 schemes, and that says which of its own cells it cannot judge. Each cell is trusted; or insufficient evidence, in which case the interval is still shown but flagged and the scheme demoted; or refused, in which case the interval is withheld. Today 8 of 17 cells are insufficient evidence and 0 are refused. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses more than four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

References

- Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2), 2023. doi: 10.1214/23-AOS2276.
- Henrik Boström and Ulf Johansson. Mondrian conformal regressors. In *Conformal and Probabilistic Prediction and Applications*, volume 128 of *Proceedings of Machine Learning Research*, pages 1–20, 2020.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- Prashansa Gupta and Sean MacAvaney. On survivorship bias in MS MARCO. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022. doi: 10.1145/3477495.3531832.
- Ying Jin and Zhimei Ren. Confidence on the focal: Conformal prediction with selection-conditional coverage. *arXiv preprint arXiv:2403.03868*, 2024.
- Tanishq Kumar, Zachary Ankner, Benjamin F. Spector, Blake Bordelon, Niklas Muennighoff, Manish Paul, Cengiz Pehlevan, Christopher Ré, and Aditi Raghunathan. Scaling laws for precision, 2024.
- Eldar Kurtić, Alexandre Marques, Mark Kurtz, and Dan Alistarh. We ran over half a million evaluations on quantized LLMs—here’s what we found. Red Hat Developer, October 2024. <https://developers.redhat.com/articles/2024/10/17/we-ran-over-half-million-evaluations-quantized-llms>.
- Jing Lei, Max G’Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523): 1094–1111, 2018. doi: 10.1080/01621459.2017.1307116.
- Yujia Tong, Yuxi Wang, Yunyang Wan, Tian Zhang, Junhao Dong, and Jingling Yuan. Does compression preserve uncertainty? a unified benchmark for quantized and sparse LLMs via conformal prediction. *arXiv preprint arXiv:2606.01850*, 2026. Wuhan University of Technology; Nanyang Technological University.
- vLLM Project. llm-compressor, 2024. <https://github.com/vllm-project/llm-compressor>.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

- 305 Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant:
306 Accurate and efficient post-training quantization for large language models. *arXiv preprint*
307 *arXiv:2211.10438*, 2022.
- 308 Yuchen Zeng and Dimitris Papailiopoulos. You don't need to run every eval. *arXiv preprint*
309 *arXiv:2606.24020*, 2026. Microsoft Research, AI Frontiers.